

BADGER, NL, CanadaWMO: **714000**Lat: **48.968N**Lon: **56.061W**Elev: **337**StdP: **14.52**Time Zone: **-3.50 (NAF)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.7	-8.0	-21.3	1.7	-13.8	-13.9	2.6	-7.4	21.2	24.1	18.9	25.0	1.6	280	0.624

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.7	83.1	65.6	79.9	64.5	76.7	63.5	69.9	78.0	67.8	74.7	65.9	72.7	8.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.0	100.7	73.6	65.3	94.6	71.7	63.2	87.9	69.5	34.1	78.2	32.4	74.8	30.9	72.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.0	15.5	13.6	-23.1	89.6	5.4	2.9	-27.0	91.6	-30.1	93.3	-33.2	94.9	-37.1	97.0		
(5)			-23.5	73.5	5.2	1.4	-27.3	74.5	-30.3	75.3	-33.3	76.0	-37.1	77.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.1	17.6	16.8	23.6	34.3	44.3	53.5	62.6	61.0	52.5	42.7	33.9	24.9
	DBStd	17.64	10.57	11.43	10.33	6.91	6.62	7.55	6.23	6.28	7.31	6.71	8.21	9.83
	HDD50	5044	1005	930	817	470	204	47	2	3	55	246	487	778
	HDD65	9536	1470	1350	1282	920	641	352	120	155	380	690	933	1243
	CDD50	1070	0	0	0	0	28	152	392	344	129	21	4	0
	CDD65	86	0	0	0	0	0	7	45	30	4	0	0	0
	CDH74	1284	0	0	0	0	31	191	605	394	62	1	0	0
	CDH80	270	0	0	0	0	6	40	136	74	14	0	0	0
Wind	WSAvg	5.6	5.9	6.0	6.4	6.3	5.9	5.4	5.0	4.6	5.1	5.2	5.6	5.8
Precipitation	PrecAvg	49.6	4.5	4.1	3.9	3.2	3.3	3.3	4.1	4.3	4.2	4.7	4.7	4.9
	PrecMax	60.0	7.4	6.7	5.9	6.5	5.2	5.6	7.5	6.9	7.3	9.1	7.8	8.8
	PrecMin	42.0	1.4	1.6	2.1	1.6	1.2	1.3	1.8	1.8	1.8	2.7	2.5	3.3
	PrecStd	4.6	1.4	1.3	1.0	1.2	0.9	1.1	1.4	1.5	1.5	1.8	1.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.1	46.8	50.7	60.8	78.2	84.5	87.7	86.3	81.0	68.3	59.7	51.0
		MCWB	48.6	44.7	42.9	49.1	56.7	63.9	68.0	67.3	65.9	60.0	56.6	49.9
	2%	DB	42.2	41.3	44.5	54.3	70.0	79.4	83.7	81.6	74.2	63.6	55.7	45.5
		MCWB	40.6	39.6	39.4	46.2	55.4	61.1	66.7	65.8	63.0	57.5	53.6	44.3
	5%	DB	36.8	36.7	40.7	50.0	64.6	74.8	80.5	78.4	70.3	59.5	51.5	41.0
		MCWB	35.6	34.7	37.0	43.4	52.1	59.9	65.5	64.5	62.0	54.3	49.7	39.7
	10%	DB	32.9	33.4	37.5	46.2	59.5	70.2	77.2	75.1	67.0	55.6	46.7	37.7
		MCWB	31.6	31.6	34.3	40.3	50.3	58.4	64.5	63.5	60.3	51.3	44.8	36.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.4	45.1	44.9	52.9	61.7	69.2	72.7	71.6	69.3	62.9	57.6	50.3
		MCDB	49.7	46.4	48.0	56.4	69.3	77.5	82.2	80.3	74.6	66.0	59.2	51.2
	2%	WB	41.0	39.9	40.6	48.2	57.4	65.3	70.6	69.2	66.3	58.8	53.8	44.5
		MCDB	42.1	41.1	43.2	52.5	64.4	72.8	79.1	77.1	71.3	62.3	55.3	45.4
	5%	WB	35.8	35.1	37.4	44.3	54.2	62.4	68.4	67.2	63.7	55.2	49.9	39.9
		MCDB	36.8	36.5	40.2	48.6	62.6	70.7	75.6	74.2	68.6	58.5	51.8	40.7
	10%	WB	31.9	31.9	34.6	41.0	51.0	59.5	66.2	65.4	61.1	51.6	44.7	36.6
		MCDB	32.8	33.2	37.0	45.5	58.7	68.0	73.7	72.2	66.2	55.1	46.4	37.5
Mean Daily Temperature Range	5% DB	MDBR	20.0	22.5	21.3	18.3	22.6	23.7	23.7	23.2	18.7	15.1	15.0	
		MCDBR	19.8	21.1	24.1	25.2	35.8	34.8	31.1	30.3	28.2	25.0	19.3	16.5
		MCWBR	19.6	19.8	19.5	17.9	21.7	19.4	16.7	17.2	18.7	19.2	18.1	16.0
	5% WB	MCDBR	19.3	19.8	21.1	22.0	29.0	26.2	23.6	22.9	23.1	22.2	18.8	16.2
		MCWBR	19.3	19.1	18.3	17.2	19.4	16.1	14.4	14.6	16.8	18.4	18.2	16.2
Clear-Sky Solar Irradiance	taub	0.299	0.308	0.340	0.367	0.365	0.383	0.411	0.393	0.363	0.332	0.309	0.300	
	taud	2.274	2.293	2.285	2.324	2.405	2.373	2.328	2.366	2.453	2.507	2.479	2.320	
	Ebn at Noon	237	266	275	279	284	279	268	268	266	257	237	220	
	Edh at Noon	23	29	35	37	36	37	38	35	29	23	19	20	
All-Sky Solar Radiation	RadAvg	285	521	895	1254	1507	1601	1628	1355	1029	630	333	209	
	RadStd	27	70	83	144	112	177	128	96	77	57	37	21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	+2.20	N/A	+0.70	+0.97	N/A	N/A	N/A	N/A

Nomenclature: See separate page